

Supplemental material Lecture 12
AA 24-25

Preliminaries

N Number of treatments

$w(d_i) \equiv w_i \rightarrow$ Output of the Simulation Model for treatment d_i (Deterministic)

$W(d) \rightarrow$ Output of the Simulation Output for treatment d (Random), including the GP contribution

$\hat{y}(d) \rightarrow$ Prediction for treatment d (Random)

The output of the Random Model (including the GP contribution) is $W(d) = \mathbf{x}(d)^T \boldsymbol{\beta} + M(d)$

Output of the Deterministic Simulation Model: $\mathbf{w}_D = (w_1, w_2, \dots, w_N)$

Random Output of the Deterministic Simulation Model: $\mathbf{W}_D = (W_1, W_2, \dots, W_N)$

The covariance structure of the Gaussian Process is described in the matrix $\mathbf{R}$ (dimensions: $N \times N$)

$$\sigma^2 \mathbf{R} = \begin{bmatrix} \sigma^2 & \text{cov}(M(d_1), M(d_2)) & \dots & \text{cov}(M(d_1), M(d_N)) \\ \text{cov}(M(d_2), M(d_1)) & \sigma^2 & \dots & \text{cov}(M(d_2), M(d_N)) \\ \dots & \dots & \dots & \dots \\ \text{cov}(M(d_N), M(d_1)) & \text{cov}(M(d_N), M(d_2)) & \dots & \sigma^2 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 1 & R_{12} & \dots & R_{1N} \\ R_{21} & 1 & \dots & R_{2N} \\ \dots & \dots & \dots & \dots \\ R_{N1} & R_{N2} & \dots & 1 \end{bmatrix} \quad \text{with } R_{ij} = \frac{\text{cov}(M(d_i), M(d_j))}{\sigma^2}$$

Moreover $\mathbf{R}^{-1}$ exists and $\mathbf{R} = \mathbf{R}^T$

The covariance between a new treatment and the known observations is described by the vector $\mathbf{r}(d)$ (dimensions: $N \times 1$):

$$\sigma^2 \mathbf{r}(d) = (\text{cov}(M(d), M(d_1)), \text{cov}(M(d), M(d_2)), \dots, \text{cov}(M(d), M(d_N)))^T$$

In the following to simplify the notation we will drop d in $\mathbf{r}$, $\mathbf{c}$ and $\mathbf{x}$

$$\mathbf{r}(d) \rightarrow \mathbf{r} \quad \mathbf{c}(d) \rightarrow \mathbf{c} \quad \mathbf{x}(d) \rightarrow \mathbf{x}$$

1. Linear Predictor

$$\hat{y}(d) = \mathbf{c}^T \mathbf{w}_D \rightarrow \text{deterministic}$$

$$\hat{y}(d) = \mathbf{c}^T \mathbf{W}_D \rightarrow \text{random}$$

2. Unbiasedness condition for the Linear Predictor

$$E(\hat{y}(d)) = E(W(d))$$

$$E(\hat{y}(d)) = E(\mathbf{c}^T \mathbf{W}_D) = \mathbf{c}^T E(\mathbf{W}_D) = \mathbf{c}^T \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{c}$$

$$E(W(d)) = \boldsymbol{\beta}^T \mathbf{x}$$

Summing up, the unbiasedness condition is: $\mathbf{X}^T \mathbf{c} = \mathbf{x}$

Note that the previous identity is more stringent than the original constraint, but this does not influence the solution for $\mathbf{c}$.

3. Development of $MSPE[\hat{y}(d)]$

We want to prove that $MSPE[\hat{y}(d)] = \sigma^2 (1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r})$

The error in the treatment d is:

$$\begin{aligned}\hat{y}(d) - W(d) &= \mathbf{c}^T \mathbf{W}_D - W(d) = \mathbf{c}^T (\mathbf{X} \boldsymbol{\beta} + \mathbf{M}_D) - (\mathbf{x}^T \boldsymbol{\beta} + M(d)) = \\ &= \mathbf{c}^T \mathbf{M}_D - M(d) + \mathbf{c}^T \mathbf{X} \boldsymbol{\beta} - \mathbf{x}^T \boldsymbol{\beta} = \mathbf{c}^T \mathbf{M}_D - M(d) + (\mathbf{c}^T \mathbf{X} - \mathbf{x}^T) \boldsymbol{\beta} = \\ &= \mathbf{c}^T \mathbf{M}_D - M(d) + (\mathbf{X}^T \mathbf{c} - \mathbf{x})^T \boldsymbol{\beta}\end{aligned}$$

But we know that the unbiasedness condition is $\mathbf{X}^T \mathbf{c} - \mathbf{x} = 0$

And the error becomes $\hat{y}(d) - W(d) = \mathbf{c}^T \mathbf{M}_D - M(d)$

A reasonable relation for the variance of the prediction is:

$$\begin{aligned}V(\hat{y}(d)) &= MSPE(d) = E((\hat{y}(d) - W(d))^2) = E((\mathbf{c}^T \mathbf{M}_D - M(d))^2) = \\ &= E(\mathbf{c}^T \mathbf{M}_D \mathbf{M}_D^T \mathbf{c} - 2\mathbf{c}^T \mathbf{M}_D M(d) + M(d)^2) = E(\mathbf{c}^T \mathbf{M}_D \mathbf{M}_D^T \mathbf{c} - 2\mathbf{c}^T \mathbf{M}_D M(d) + M(d)^2)\end{aligned}$$

Considering that:

- 1) $E(\mathbf{c}^T \mathbf{M}_D \mathbf{M}_D^T \mathbf{c}) = \sigma^2 \mathbf{c}^T \mathbf{R} \mathbf{c}$
- 2) $E(2\mathbf{c}^T \mathbf{M}_D M(d)) = 2\mathbf{c}^T \mathbf{r}$
- 3) $E(M(d)^2) = \sigma^2$

We have $MSPE(d) = \sigma^2 (1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r})$

4. Development of the Linear Predictor

We have to solve the following constrained optimization problem:

$$\begin{aligned}\min_{\mathbf{c}} MSPE[\hat{y}(d)] &= \sigma^2 (1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r}) \\ \text{s.t. } \mathbf{X}^T \mathbf{c} &= \mathbf{x} \quad \text{unbiasedness condition}\end{aligned}$$

We use the Lagrange multiplier method to solve the optimization problem.

The Lagrangian is:

$$L(\mathbf{c}, \boldsymbol{\lambda}) = (1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r}) + \boldsymbol{\lambda}^T (\mathbf{X}^T \mathbf{c} - \mathbf{x})$$

Dimension Check:

$$1*1 = 1*1 + (1*N)*(N*N)*(N*1) + (1*p)[(p*N)(N*1) - (p*1)] \quad \text{OK}$$

$$L(\mathbf{c}, \boldsymbol{\lambda}) = 1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r} + \boldsymbol{\lambda}^T \mathbf{X}^T \mathbf{c} - \boldsymbol{\lambda}^T \mathbf{x}$$

Or (note that the addition terms are scalars and the transpose does not affect the scalar)

$$L(\mathbf{c}, \boldsymbol{\lambda}) = 1 + \mathbf{c}^T \mathbf{R} \mathbf{c} - 2\mathbf{c}^T \mathbf{r} + \mathbf{c}^T \mathbf{X} \boldsymbol{\lambda} - \boldsymbol{\lambda}^T \mathbf{x}$$

The necessary conditions are;

$$\frac{\partial L}{\partial \mathbf{c}} = \mathbf{0} \rightarrow 2\mathbf{R}\mathbf{c} - 2\mathbf{r} + \mathbf{X}\boldsymbol{\lambda} = \mathbf{0}$$

$$\frac{\partial L}{\partial \boldsymbol{\lambda}} = \mathbf{0} \rightarrow \mathbf{X}^T \mathbf{c} - \mathbf{x} = \mathbf{0}$$

We can write the two equations as:
$$\begin{bmatrix} \mathbf{0} & \mathbf{X}^T \\ \mathbf{X} & 2\mathbf{R} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \mathbf{c} \end{bmatrix} = \begin{bmatrix} \mathbf{x} \\ 2\mathbf{r} \end{bmatrix}$$

or
$$\begin{bmatrix} 2\mathbf{R} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2\mathbf{r} \\ \mathbf{x} \end{bmatrix}$$

If we invert the Square Matrix, we have:
$$\begin{bmatrix} \boldsymbol{\lambda} \\ \mathbf{c} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{X}^T \\ \mathbf{X} & 2\mathbf{R} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{x} \\ 2\mathbf{r} \end{bmatrix}$$

Or
$$\begin{bmatrix} \mathbf{c} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2\mathbf{R} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} 2\mathbf{r} \\ \mathbf{x} \end{bmatrix}$$

Now we have to compute the inverse of a partitioned matrix (i.e. see [Searle \[1\] pg 268 Exercise 13c](#))

If $\mathbf{H} = \begin{bmatrix} \mathbf{0} & \mathbf{P} \\ \mathbf{Q} & \mathbf{A} \end{bmatrix} \rightarrow \mathbf{H}^{-1} = \begin{bmatrix} -(\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} & (\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \mathbf{P}\mathbf{A}^{-1} \\ \mathbf{A}^{-1}\mathbf{Q}(\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} & \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{Q}(\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \mathbf{P}\mathbf{A}^{-1} \end{bmatrix}$

Using our symbols, we have:

$$\mathbf{P} \rightarrow \mathbf{X}^T$$

$$\mathbf{Q} \rightarrow \mathbf{X}$$

$$\mathbf{A} \rightarrow 2\mathbf{R} \text{ and } \mathbf{A}^{-1} \rightarrow \frac{1}{2}\mathbf{R}^{-1}$$

then

$$-(\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \rightarrow -2(\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1}$$

$$(\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \mathbf{P}\mathbf{A}^{-1} \rightarrow 2(\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \frac{1}{2} \mathbf{R}^{-1} = (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}^{-1}$$

$$\mathbf{A}^{-1} \mathbf{Q} (\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \rightarrow \frac{1}{2} \mathbf{R}^{-1} \mathbf{X} 2(\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1}$$

$$\mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{Q} (\mathbf{P}\mathbf{A}^{-1}\mathbf{Q})^{-1} \mathbf{P}\mathbf{A}^{-1} \rightarrow \frac{1}{2} \mathbf{R}^{-1} - \frac{1}{2} \mathbf{R}^{-1} \mathbf{X} 2(\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \frac{1}{2} \mathbf{R}^{-1} = \frac{1}{2} \mathbf{R}^{-1} - \frac{1}{2} \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}^{-1}$$

If we substitute in the equality and multiply, we obtain the vector $\mathbf{c}$

$$\mathbf{c} = \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{x} + \left(\frac{1}{2} \mathbf{R}^{-1} - \frac{1}{2} \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}^{-1} \right) 2\mathbf{r} =$$

$$= \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{x} + \mathbf{R}^{-1} \mathbf{r} - \mathbf{R}^{-1} \mathbf{X} (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}^{-1} \mathbf{r}$$

Or

$$MSPE = V[\hat{y}(d)] = \sigma^2 \left(1 - \begin{bmatrix} \mathbf{x}(d)^T & \mathbf{r}(d)^T \end{bmatrix} \begin{bmatrix} 0 & \mathbf{X}^T \\ \mathbf{X} & \mathbf{R} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{x}(d) \\ \mathbf{r}(d) \end{bmatrix} \right)$$

Try to prove the last formula

6. MLE Maximum Likelihood Estimator

Previous formulas do not consider the problem of estimating $\mathbf{R}$, $\mathbf{r}$ and the process variance σ^2 . To solve this problem we will assume:

- 1) A single factor (treatment d with one dimension)
- 2) Exponential weighting scheme (p and the single parameter θ)
- 3) MLE method and we will use the approach “divide and conquer”, called “profile likelihood” or “concentrated likelihood” in mathematical statistics. In other words first we will estimate the regression parameters and the process variance, then we will estimate the parameter θ and p

By assumptions, the observation sample (Training set) follows a Multivariate Normal Density function:

$$\mathbf{W}_D : \{W(d_1), W(d_2), \dots, W(d_N)\} \sim N_N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{R}(\theta, p))$$

The likelihood function is the Multivariate Normal Density function computed in the training set values.

The Likelihood expression (to be maximized) is:

$$L(\boldsymbol{\beta}, \sigma^2, \theta, p) = (2\pi)^{-N/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})\right)$$

We usually prefer to maximize the **Log Likelihood function** which is:

$$l(\boldsymbol{\beta}, \sigma^2, \theta, p) = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log \sigma^2 - \frac{1}{2} \log |\mathbf{R}(\theta, p)| - \frac{1}{2\sigma^2} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})^T \mathbf{R}^{-1}(\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})$$

Since $\boldsymbol{\Sigma} = \sigma^2 \mathbf{R}(\theta, p)$ and $|\sigma^2 \mathbf{R}(\theta, p)| = (\sigma^2)^N |\mathbf{R}(\theta, p)|$

Or we **minimize** $-l(\boldsymbol{\beta}, \sigma^2, \theta, p)$:

$$-l(\boldsymbol{\beta}, \sigma^2, \theta, p) = \frac{N}{2} \log(2\pi) + \frac{N}{2} \log \sigma^2 + \frac{1}{2} \log |\mathbf{R}(\theta, p)| + \frac{1}{2\sigma^2} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})^T \mathbf{R}^{-1}(\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})$$

The necessary condition for an extremum is:

$$\frac{\partial(-l(\boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}, p))}{\partial \sigma^2} = \frac{N}{2\sigma^2} - \frac{1}{2(\sigma^2)^2} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})^T \mathbf{R}^{-1} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta}) = 0$$

$$\hat{\sigma}^2 = \frac{1}{N} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})^T \mathbf{R}^{-1} (\mathbf{W}_D - \mathbf{X}\boldsymbol{\beta})$$

$$\begin{aligned} \frac{\partial(-l(\boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}, p))}{\partial \boldsymbol{\beta}} &= \frac{\partial}{\partial \boldsymbol{\beta}} \left[\frac{1}{2\sigma^2} (\mathbf{W}_D^T \mathbf{R}^{-1} \mathbf{W}_D - \mathbf{W}_D^T \mathbf{R}^{-1} \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{R}^{-1} \mathbf{W}_D + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{R}^{-1} \mathbf{X}\boldsymbol{\beta}) \right] = \\ &= \frac{\partial}{\partial \boldsymbol{\beta}} \left[\frac{1}{2\sigma^2} (\mathbf{W}_D^T \mathbf{R}^{-1} \mathbf{W}_D - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{R}^{-1} \mathbf{W}_D + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{R}^{-1} \mathbf{X}\boldsymbol{\beta}) \right] = \frac{1}{2\sigma^2} (-2\mathbf{X}^T \mathbf{R}^{-1} \mathbf{W}_D + 2\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X}\boldsymbol{\beta}) = 0 \\ (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X}) \boldsymbol{\beta}^* &= \mathbf{X}^T \mathbf{R}^{-1} \mathbf{W}_D \\ \boldsymbol{\beta}^* &= (\mathbf{X}^T \mathbf{R}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}^{-1} \mathbf{W}_D \end{aligned}$$

The two formulas for the first step of the optimization algorithm are:

Regression parameters: $\boldsymbol{\beta}^* = (\mathbf{X}^T \mathbf{R}(\boldsymbol{\theta}, p)^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}(\boldsymbol{\theta}, p)^{-1} \mathbf{w}_D$

Note:

- we substitute $\mathbf{W}_D \rightarrow \mathbf{w}_D$ to allow the computations
- The regression parameters depend on $\boldsymbol{\theta}$ and $p \rightarrow \boldsymbol{\beta}^*(\boldsymbol{\theta}, p)$
- we assume that the matrix $(\mathbf{X}^T \mathbf{R}(\boldsymbol{\theta}, p)^{-1} \mathbf{X})$ is invertible
- We put the star because we must go to the second step to have a complete estimator since we do not know the weight parameters.

Process variance equation: $\sigma^* = \left(\frac{1}{N} (\mathbf{w}_D - \mathbf{X}\boldsymbol{\beta}^*)^T \mathbf{R}(\boldsymbol{\theta}, p)^{-1} (\mathbf{w}_D - \mathbf{X}\boldsymbol{\beta}^*) \right)^{1/2}$

Note:

- we substitute $\mathbf{W}_D \rightarrow \mathbf{w}_D$ to allow the computations
- we divide by N, but it is better to divide by $(N - df_E)$

After the first step we have two expressions which depend on the weight parameters $\boldsymbol{\theta}$ and p . We substitute the two equations in the Likelihood or LogLikelihood function and we numerically optimize and obtain estimates of the weight parameters $\hat{\boldsymbol{\theta}}$ and $\hat{p}$.

References

[1] Shayle R. Searle "Matrix Algebra useful for Statistics" John Wiley & Sons, 1982